SWIFT-U5 — Final Methodological Gate Report

Date: 2026-11-08

Repository: `/Users/rezaul1424/Everything/SWIFT-U5`

Branch: `main`

Test Suite: 62/62 passing

Smoke Test: 16/16 stages verified

Phase 6 Status: [GATE CLEARED] — AWAITING EXPLICIT USER AUTHORIZATION

Executive Summary

Prior to launching the full Phase 6 execution, this final methodological audit examined five critical areas:

1. Module A Weight Sensitivity: Evaluated four composite clipping levels (Unclipped, 95th, 99th primary, 99.5th percentile). Confirmed that the prespecified 99th-percentile clip raises effective sample size from 3,930 (13.2%) to 6,190 (20.8%) on development data while preserving development weighted prevalence (0.0498 vs 0.0497) and maintaining validation discrimination (ΔAUROC ≤ 0.013). The prespecified 99th percentile is retained as primary.

2. Threshold Provenance in Rolling-Origin Experiments: Uncovered that the previous `last training round (val proxy)` implementation evaluated in-sample predictions for temporal cuts. Implemented a principled two-pass out-of-sample architecture (`_derive_oos_threshold_temporal`):

- For multi-round cuts (k ≥ 2), a threshold model is trained on rounds [r_1, \dots, r_k-1] and evaluated on r_k (genuinely out-of-sample).
- For single-round cuts (k = 1), 5-fold stratified cross-validation produces out-of-fold predictions.
- For random arms, a 3-way split (70% train / 15% threshold val / 15% test) guarantees zero label leakage.
- The primary 2022 deployment evaluation remains strictly locked to the held-out 2017–18 validation round.

3. Weighted vs. Unweighted Threshold Analysis: Investigated the identical unrounded values (e.g., 0.048556809056) observed in validation. Confirmed that both weighted and unweighted Youden indices (J = TPR - FPR) peaked at the exact same discrete predicted probability point on the validation set, though their ROC curves comprised 7,516 and 1,592 thresholds respectively.

4. Bootstrap Replicate & Metadata Tracking: Upgraded `bootstrap_metrics()` to record replicate counts (`n_requested`, `n_succeeded`, `n_failed_degenerate`) and explicitly log excluded non-numeric metadata (`threshold_source`). Also wired frozen validation thresholds directly through the bootstrap loop to evaluate fixed decision rules across resamples.

5. Call-Site Audit: Verified every invocation of `compute_all_metrics()` across the codebase. Confirmed that all production test evaluations supply frozen external thresholds and record `threshold_source = 'external_validation'`.

1. Module A Weight Concentration & Sensitivity Analysis

1.1 Multi-Level Clipping Comparison

Using the permitted development cohort (2004–2014, N = 29,686) with shift weighting target set to the 2017–18 validation round, we evaluated four composite weight clipping regimes:

Configuration	W_max	W_std	N_clipped	\%_clipped	ESS_pre	ESS_post	ESS / N	Wgt Prev	Val AUROC (W)	Val AUROC (UW)	Δ AUROC_W
Unclipped (Original)	170.76	2.5600	0	0.0%	3,930	3,930	13.2%	0.0497	0.6478	0.6563	+0.0033
95th Percentile	6.14	1.4664	1,485	5.0%	3,930	9,423	31.7%	0.0503	0.6560	0.6661	+0.0115
99th Percentile [PRIMARY]	12.29	1.9482	297	1.0%	3,930	6,190	20.8%	0.0498	0.6445	0.6546	0.0000
99.5th Percentile	15.39	2.0621	149	0.5%	3,930	5,652	19.0%	0.0499	0.6578	0.6661	+0.0133

Notes:

- ESS is computed via Kish's formula: (Σ W_i)^2 / Σ W_i^2.
- The 99th-percentile clip drops the weight ceiling by 92.8% (from 170.76 down to 12.29) while altering only 1.0% of records.
- Development weighted prevalence remains stable (0.0497 → 0.0498).
- Predictive performance on held-out 2017–18 data is stable across all configurations (maximum variation Δ = 0.0133).

1.2 Wave-Level Stability (Primary 99th-Percentile Configuration)

Inspection of effective sample size across survey waves under the primary configuration demonstrates robust representation:

Survey Wave	Nominal N	Weight Max	Wave ESS	Wave ESS / Nominal
2004	6,908	12.29	3,369	48.8%
2007	6,150	12.29	3,767	61.3%
2011	8,742	3.26	6,762	77.3%
2014	7,886	12.29	3,039	38.5%

Survey Wave	Nominal N	Weight Max	Wave ESS	Wave ESS / Nominal
Total Dev	29,686	12.29	6,190	20.8%

Methodological Conclusion:

The prespecified 99th-percentile composite clip successfully curbs extreme density-ratio multipliers without distorting population prevalence or wave-level representation. In accordance with the study plan, it is retained as the primary specification, and the unclipped regime will be reported as a sensitivity analysis in manuscript tables.

2. Threshold Provenance in Every Temporal Experiment

2.1 Audit of the Rolling-Origin Sequence

The rolling-origin benchmark (`src/validation/experiment_2x2.py`) evaluates 5 sequential temporal transitions. The audit revealed an issue in the earlier draft: using the "last training round" as a validation proxy meant the model evaluating those records had already seen them during training (in-sample bias of Δ ≈ 0.012 on threshold).

This was corrected by implementing a two-pass out-of-sample threshold generator:

Threshold Model fitted on \r_1, \dots, r_k-1\ → Predict on r_k \ (OOS) → Freeze τ_Youden

Cut	Training Rounds (N_train)	Test Round (N_test)	Threshold Derivation Source	Sample Size	Out-of-Sample Status	Threshold Frozen Before Test?
C1	2004 (6,908)	2007 (6,150)	5-Fold Stratified CV on 2004	6,908	Out-of-Fold	YES
C2	2004, 2007 (13,058)	2011 (8,742)	Model on 2004 → Predict 2007	6,150	Out-of-Sample	YES
C3	2004–2011 (21,800)	2014 (7,886)	Model on 2004–07 → Predict 2011	8,742	Out-of-Sample	YES
C4	2004–2014 (29,686)	2017–18 (8,684)	Model on 2004–11 → Predict 2014	7,886	Out-of-Sample	YES
C5	2004–2017-18 (38,370)	2022 (13,119)	Model on 2004–14 → Predict 2017–18	8,684	Out-of-Sample	YES

2.2 Random Split Arms (Arms A & B)

For the random split arms, a 3-way split of the training pool was instituted:

- Model Training: 70% of pooled observations
- Threshold Optimization: 15% held-out validation subset (never seen by threshold model)
- Test Evaluation: 15% held-out test subset (evaluated using frozen threshold)

2.3 Primary 2022 Deployment Evaluation

The primary research paper deployment cut (Table 4 & Table 5):

- Development Data: 2004, 2007, 2011, 2014 (N = 29,686)
- Validation Holdout: 2017–18 (N = 8,684) — supplies both Module C drift calibration parameters and frozen Youden thresholds
- Locked Test Round: 2022 (N = 13,119) — evaluated strictly out-of-sample with zero parameter adaptation

3. Investigation of Weighted vs. Unweighted Threshold Values

In the Phase 5 smoke test, both weighted and unweighted thresholds for the 2017–18 validation set were reported as `0.04546`. An in-depth numerical inspection was conducted:

1. ROC Curve Threshold Counts:
   - Unweighted ROC curve: 1,592 thresholds
   - Weighted ROC curve: 7,516 thresholds
2. Youden Index Optimization:
   - Unweighted Youden index: max(J) = 0.277348 at Index 590
   - Weighted Youden index: max(J) = 0.260085 at Index 1677
3. Exact Threshold Equality:
   - τ_unweighted = 0.048556809056 (unrounded)
   - τ_weighted = 0.048556809056 (unrounded)
   - Absolute difference: 0.000000000000

Mathematical Explanation:

In `scikit-learn`, `roc_curve()` constructs candidate decision thresholds directly from the sorted unique values in `y_prob`. In low-prevalence epidemiology cohorts (P ≈ 4\%), the local maximum of the objective function J(τ) = Sensitivity(τ) + Specificity(τ) - 1 occurs along an empirical probability boundary where positive cases concentrate. Because survey sample weights re-weight observations rather than altering predicted values, the maximum of J_W and J_UW coincided on the exact same discrete predicted probability point in the validation cohort. When models differ (e.g. baseline XGBoost vs full SWIFT-U5), the weighted and unweighted thresholds diverge (e.g., 0.3836 vs 0.4356, Δ = 0.052).

4. Audit of All `compute_all_metrics()` Call Sites

Every call to `compute_all_metrics()` was audited across the repository:

File & Line	Context	Thresholds Passed?	Output	Status
`experiments/phase5_smoke_test.py:367`	Locked 2022 Test	τ_W, τ_UW from 2017–18 val	`external_validation`	VERIFIED
`experiments/04_run_swift_u5_ablation.py:131`	Locked 2022 Test	τ_W, τ_UW from 2017–18 val	`external_validation`	VERIFIED
`src/validation/experiment_2x2.py:186`	Rolling-Origin Tests	τ_W, τ_UW from OOS generator	`external_validation`	VERIFIED
`src/evaluation/bootstrap.py:125`	Bootstrap Replicates	τ_W, τ_UW from caller	Evaluates frozen τ	VERIFIED
`tests/test_pre_phase6_verification.py`	Unit Tests	Explicitly tested both paths	`external_validation` & `internal`	VERIFIED

Accidental Optimization Prevention:

In all test-evaluation workflows, missing thresholds trigger explicit warning metadata (`threshold_source = 'internal_from_passed_data'`), which automated pipeline integrity assertions reject.

5. Bootstrap Reporting Verification

`src/evaluation/bootstrap.py` was updated to provide full transparency into resampling integrity:

1. Replicate Accounting:
   - `_bootstrap_meta.n_requested`: Total draws scheduled (1,000 for benchmark)
   - `_bootstrap_meta.n_succeeded`: Successfully evaluated resamples
   - `_bootstrap_meta.n_failed_degenerate`: Draws where all records are un-diseased or all diseased (recorded, never silently suppressed)
2. Metadata Filtering:
   - Non-numeric string columns (such as `threshold_source`) are identified, excluded from numerical summary calculations, and explicitly listed under `_bootstrap_meta.non_numeric_fields_excluded`.
3. Smoke Test Bootstrap Verification (10 Replicates):
   - Replicates requested: 10
   - Replicates succeeded: 10
   - Replicates degenerate: 0
   - Excluded metadata: `['threshold_source']`
   - Numeric metrics returned: 33 (AUROC, AUPRC, Brier, ECE, Sensitivity, PPV, NNS, etc.)

6. Verification Test Suite Results

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Ran 62 tests in 0.672s — OK (ALL PASSING)
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Test Class	Focus Area	Tests	Status
`TestModuleAWeightDiagnostics`	Composite clipping, ESS calculation, weight normalization	5	[PASS]
`TestYoudenThresholdProvenance`	Validation threshold freezing, external parameter enforcement	6	[PASS]
`TestAblationRunnerCompatibility`	Split disjointness, N_boot=1000, `df_val` calibration	4	[PASS]
`TestOOSThresholdDerivation`	Penultimate-round OOS model, CV fallback, test disjointness	5	[PASS]
`TestBootstrapMetaReporting`	Replicate counters, non-numeric field logging	4	[PASS]
`TestPhase3Corrections`	PSU wave-scoping, right-censoring flags, b5/b7 leakage exclusion	38	[PASS]
Total	Comprehensive Pre-Phase 6 Test Suite	62	[100% PASS]

7. Remaining Methodological Risks & Handling

Risk Description	Severity	Protocol Mitigation
Survey era distributional shift	Low	Managed via Module A recency time-decay and clipped density ratios.
Right-censoring of living children <60m	Low	Retrospective cohort estimand explicitly defined as recorded under-five death by survey interview, with follow-up distributions documented.
Cluster bootstrap compute load (1,000 reps)	Medium	NumPy vectorized indexing implemented in `bootstrap.py`; `compute_calibration=False` enabled during resampling for a ≈ 100× speedup.
Model comparison fairness	Nil	All baselines and SWIFT-U5 configurations share the identical feature matrix, temporal partitions, and frozen out-of-sample decision thresholds.

8. Final Phase 6 Readiness Verdict

GOVERNANCE DIRECTIVE / VERDICT: [GATE CLEARED]: All methodological requirements have been rigorously satisfied: - Out-of-sample threshold provenance is verified across all rolling-origin cuts. - Module A sensitivity analysis confirms the defensibility of the 99th-percentile primary clip. - The 62-test automated suite passes completely. - The end-to-end smoke test completed with zero errors or warnings. Execution is halted. Standing by for explicit authorization to begin Phase 6.